

Michael Kelly

Web Front End/Full Stack Developer

Who am I?

I am an Subject Matter Expert Software Engineer with over two decades of experience designing and building software in the intelligence community. I love to make web applications that are polished, standards based, and user friendly. While I prefer to work on the front end, my Computer Science background lends well when I need to get my hands dirty on the back end of systems. These skills have played a large part in the launches of 8 major web applications, four of them flagship level.

Contact Info

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Experience

SME Software Engineer, ByteRatio, Reston, VA (2019-Present)

Led development of an AI-powered, multi-tenant talent management platform built with Next.js 16, TypeScript, and PostgreSQL, enabling organizations to intelligently match candidates to job postings. Architected a two-stage matching engine combining OpenAI vector embeddings with GPT-5 semantic reranking, backed by an event-driven ETL pipeline across 7 AWS Lambda functions for automated resume ingestion and profile extraction. Implemented organization-scoped data isolation, role-based access control, and Microsoft Azure AD authentication to support a secure, multi-organization SaaS architecture. (2025-present)

Full Stack Developer on graph and data correlation web application using big data and developed with Vite and React for the front end. Developed Python Lambda functions and championed scalable asynchronous search features. Developed Playwright CI/CD integration using Kubernetes and Jenkins. (2024-2026)

UI Developer Lead for next generation flagship web application for entitlement management using Next.js and React. Responsible for UX design, architecting CI/CD automation, and guiding team members in best practices. (2022-2024)

Developer Lead for a web application providing search capabilities for a large datastore using Angular 11. Responsible for mentoring junior team members and ensuring best practices are followed. (2021-2022)

Front End Team Lead for a flagship web application delivering timely information for worldwide customers using Angular 8. Responsible for mentoring junior developers on the team in best practices. (2020-2021)

UI Team Lead responsible for modernizing legacy Angular/Node password management application using React. Architected repeatable deploy process using Docker and AWS. (2019-2020)

Skills and tools used:

HTML, CSS, Next.js, Vite, React JS, Angular 8,11, NodeJS, Python, TypeScript, PostgreSQL, Prisma, OpenAI, Git, Docker, Kubernetes, AWS, MCP, and RAG.

Expert Software Engineer, Excivity, Inc, Chantilly, VA (2017-2019)

Web UI Team Lead responsible for growing junior developers on team.

Designed, Architected, and built Single Page Application (SPA) for User Communications (both text based and real-time voice with historical playback of each voice message), and User Management of system

Architected and built SPA for management and interactive use (via VNC within a web page) of Android Mobile Device Emulators running in a cloud-based environment.

Researched and developed automatic build pipeline using Docker and Gitlab for release management.

Skills and tools used:

HTML, CSS, React JS, WebRTC, JavaScript, Photoshop, Git, Jira, Docker, and GitLab.

Owner and Developer, URL Droplet, Ashburn, VA (2011-2018)

Designed and developed web application for downloading files to a Dropbox account with just a URL. URL Droplet has been featured on Lifehacker (<http://lifehacker.com/5811317/url-droplet-downloads-files-from-a-web-address-straight-to-your-dropbox>), CNET (http://howto.cnet.com/8301-11310_39-20073094-285/4-killer-dropbox-tricks/), and many other news sites and blogs around the world.

Skills and tools used:

Git, Photoshop, HTML, CSS, JQuery, JavaScript, Ruby on Rails, and MongoDB.

Senior Software Engineer, GSM Consulting, Reston, VA (2012-2017)

Designed and developed a Ruby on Rails replacement for a Lotus Notes database. Transformed workflow to save 30+ man-hours/week. (2013-2017)

User Interface Design Team Lead responsible for branding and design of user experience. Consultant on multiple projects at client location. (2012-2013)

Skills and tools used:

Ruby on Rails, MySQL, HTML, CSS, JQuery, Ember, React JS, JavaScript, Photoshop, and Git.

Senior Software Engineer, NCI Inc, Reston, VA (2008-2012)

Lead Designer and Lead Developer for senior policy leadership's flagship publication web application using HTML/CSS/JS (JQuery) and Ruby on Rails. (2011-2012)

User Experience Designer for new search capability on flagship publication web application. Conducted User interviews, created prototypes and developed beta implementation. (2011)

Lead Designer and Developer for a streaming video web application using HTML/CSS/JS (Prototype) and Ruby on Rails. (2010)

Co-designed, and implemented a complete rebranding of client's flagship publication web application using HTML/CSS/JS (Prototype) and Ruby on Rails. (2009)

Developer for a publication web application using HTML/CSS/JS (Prototype) and Ruby on Rails. Responsible for designing admin tools and pages on the site using extensive use of AJAX. (2008)

Skills and tools used:

Ruby on Rails, HTML, CSS, JQuery, Prototype, Backbone, JavaScript, Photoshop, Oracle, and Git.

Lead Software Engineer, SpecTal Consultants, Reston, VA (2006-2007)

Led and managed software development team that included three separate subcontractors designing and building a multi-modal biometric software system.

Responsible for planning the development schedule as well as keeping the team on schedule and on budget.

Skills and tools used:

Microsoft Project, Java, J2EE, HTML, CSS, and Subversion.

Senior Software Engineer, L-3 Communications Titan Group, Reston, VA (2005-2006)

Led the User Interface team for a large scale Internet exploitation tool working on designs for the Graphical User Interface and integrated vendor components.

Designed and developed a Java Eclipse Plug-in for search within a knowledge management tool.

Designed and developed Java infrastructure applications to integrate a message feed into the overall architecture.

Skills and tools used:

Java

Senior Software Engineer, L-3 Communications Titan Group, Reston, VA (2004-2005)

Gathered customer requirements from the customer.

Designed and developed a business workflow processing system using Tibco.

Designed and developed Java remote execution and status display applications using JMS.

Skills and tools used:

Java, JMS, and Tibco.

Software Engineer, L-3 Communications Titan Group, Reston, VA (2002-2004)

Developed subsystems for XML message processing and correlation using C++ and Java.

Skills and tools used:

C++, Java, XML, CVS, and CMMI Level 3.

Intern, Integrity Applications Inc, Chantilly, VA (2001-2001)

Implemented a browser-based graphical analysis subsystem driven by XML data using JavaScript, VBScript, XSL, and HTML.

Skills and tools used:

HTML, CSS, JavaScript, VBScript, XML, and XSL.

Intern, Fairfax Imaging, Inc, Chantilly, VA (2000-2001)

Designed and implemented a graphical interface on top of an existing legacy application using C++ and MFC.

Skills and tools used:

C++, and MFC.

Clearance

Active TS/SCI Clearance with Polygraph

Education

B.S, Computer Science, James Madison University (2002)